| Name of the Student | G.NAGA MANIKANTA |
|---|---|
| Reg. No | 192425310 |
| GitHub Link | |

**SIMATS ENGINEERING**

**CO4 - ASSESSMENT TOOL 2**

# Mock Interview

**COURSE NAME: Deep Learning** **COURSE CODE: MLA04**
**SLOT :**

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**COURSE OUTCOME COVERED**

**CO 5:** Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

**Course Outcome Weightage: 35%** **Duration: 90 minutes**
**Assessment Tool Weightage: 20%** **Mark : 25**

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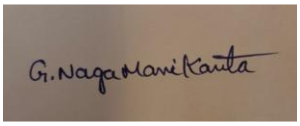
**Code of Conduct:**
I G.NAGA MANIKANTA(192425310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.




**G.MANIKANTA**

**Signature of the student:** **Name of the student:**

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| **Criteria** | Subject Knowledge and Conceptual Understanding (3) | Technical Accuracy and Application of Concepts (2.5) | Problem-Solving and Analytical Thinking **(2)** | Communication Skills and Clarity of Explanation **(1.5)** | Confidence, Presentation and Professional Behaviour ***(1)*** | **Total (10)** |
|---|---|---|---|---|---|---|
| **Weightage** | ***30 %*** | ***25%*** | ***20%*** | ***15 %*** | ***10 %*** | **100%** |

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|---|---|---|---|---|---|---|
| **Q1** | | | | | | |
| **Q2** | | | | | | |
| **Q3** | | | | | | |
| **Q4** | | | | | | |
| **Q5** | | | | | | |
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| **Q22** | | | | | | |
| **Q23** | | | | | | |
| **Q24** | | | | | | |
| **Q25** | | | | | | |

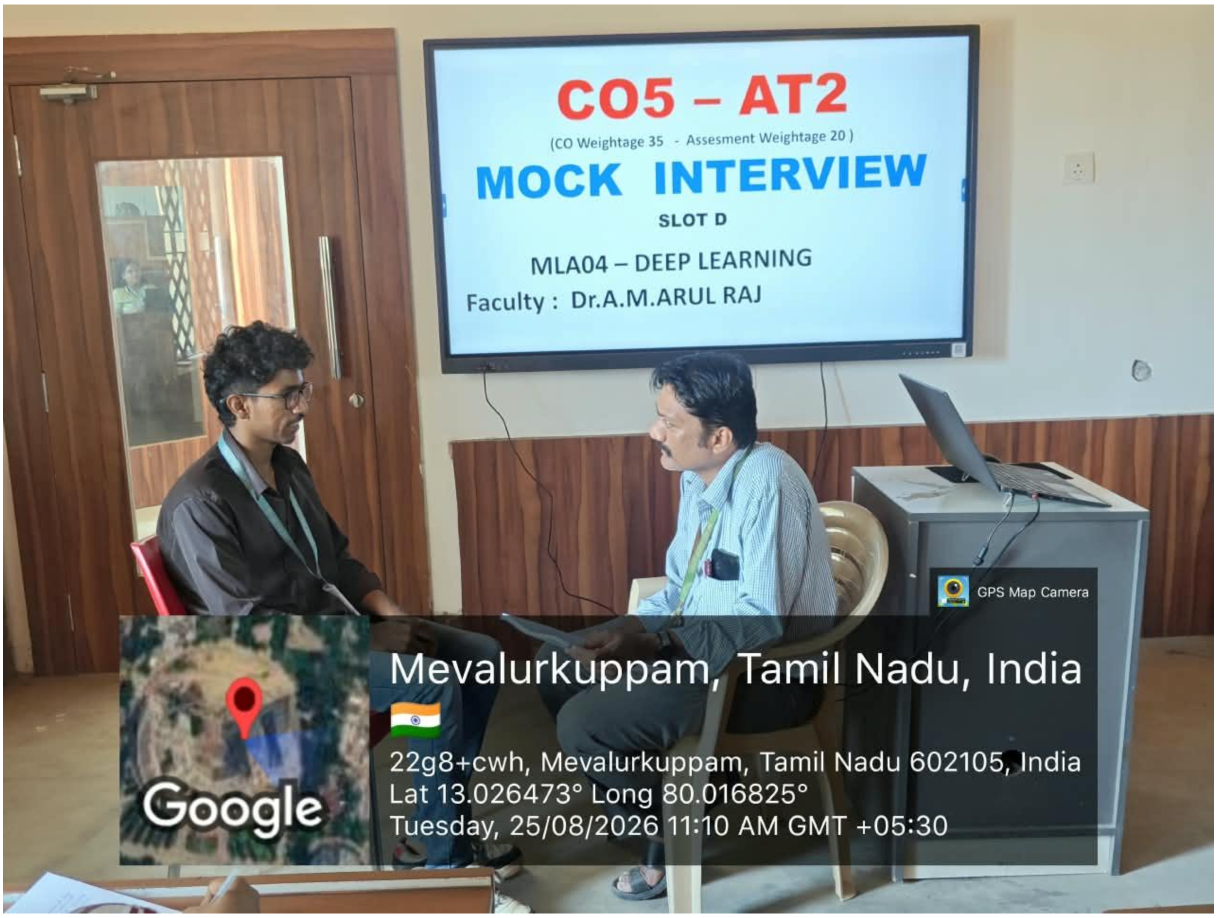
**Signature of the Faculty:** **Total Marks:**

**QUESTIONS: 25** **Total Marks:25**

1.What is an Autoencoder in Deep Learning?
2.What is the main purpose of an Autoencoder?
3.What are the major components of an Autoencoder?
4.What is the role of the Encoder in an Autoencoder?
5.What is the role of the Decoder in an Autoencoder?
6.What is meant by the Bottleneck layer in an Autoencoder?
7.What is meant by Reconstruction in an Autoencoder?
8.How does an Autoencoder learn to reconstruct its input?

9.What is the difference between an Autoencoder and a traditional Neural Network?
10.What type of loss function is commonly used to train an Autoencoder?
11.Explain the basic architecture of an Autoencoder with a suitable example.
12.Why is the dimensionality of the bottleneck layer usually smaller than the input layer?
13.What happens if the bottleneck layer has too many neurons?
14.What happens if the bottleneck layer has very few neurons?
15.What is a Denoising Autoencoder?
16.How can an Autoencoder be used for image denoising?
17.How can Autoencoders be used for dimensionality reduction?
18.How can an Autoencoder be used for anomaly detection?
19.What is the difference between a Denoising Autoencoder and a basic Autoencoder?
20.Give some real-world applications of Autoencoders.
21.What is a Deep Generative Model?
22.What is the difference between a generative model and a discriminative model?
23.What is a Variational Autoencoder (VAE)?
24.What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?
25.What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?

PHOTO:



**RUBRICS FOR CALCULATION:**

| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
|---|---|---|---|---|---|
| Subject Knowledge and Conceptual Understanding | 30% | Comprehensive understanding of real-world problem and constraints | Good understanding with minor gaps in requirements | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Technical Accuracy and Application of Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem | Applies appropriate concepts with minor errors | Limited application of concepts | Incorrect or no application of concepts |
| Problem-Solving and Analytical Thinking | 20% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps | Basic solution with limited feasibility | Unclear or impractical solution |
| Communication Skills and Clarity of Explanation | 15% | Effective use of modern tools, technologies, and real data | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools | Minimal or incorrect use of tools |
| Confidence, Presentation and Professional Behaviour | 10% | Thorough analysis with validated results and performance evaluation | Good analysis with reasonable validation | Basic analysis with limited validation | Poor or no analysis |